

Testing Industrial Policymakers:
Evidence from Brazil's BNDES
First-Stage Pipeline & Aggregation Diagnostics

March 25, 2026

Question: Is the sector-level allocation of BNDES lending across municipalities GDP-optimal?

Three-step identification:

- ① **Firm first stage** (micro validation): political alignment → firm BNDES access
- ② **Sector aggregation:** firm instruments → sector-level credit reallocation
- ③ **Municipality second stage:** predicted reallocation → GDP per capita

This presentation: steps 1 and 2 — establishing (and diagnosing) the first stage

- Firm-level signal is **present**: alignment predicts BNDES access
- Sector-level aggregation **destroys** the signal — we diagnose why
- Forward paths: coarser sectors, firm-level reallocation summaries, ungrouped CNAE

Firm-Level Specification

Unit: firm $f \times$ municipality $m \times$ year t ($N = 39.5\text{M}$, 2002–2017)

Dependent variable: $1(\text{BNDES}_{fmt} > 0)$ — did firm f receive any BNDES credit in muni m , year t ? (extensive margin, LPM)

Instruments — “Fraction Aligned” at government tier $\ell \in \{\text{Mayor, Governor, President}\}$:

$$\text{FA}_{fmt}^{\ell} = \sum_p \omega_{fp,0} \cdot \text{Align}_{mpt}^{\ell}$$

- $\omega_{fp,0}$: share of firm f 's owners affiliated with party p , averaged over the 4-year pre-election window (baseline exposure)
- $\text{Align}_{mpt}^{\ell}$: = 1 if party p is in the governing coalition at tier ℓ in muni m , year t
- FA^{ℓ} : fraction of the firm's politically affiliated owners whose party is currently in power at tier ℓ

Specification:

$$1(\text{BNDES}_{fmt} > 0) = \sum_{\ell} \beta_{\ell} \text{FA}_{fmt}^{\ell} + \phi_f + \alpha_{mt} + \varepsilon_{fmt}$$

FE: firm + muni $\times$ year. Clustering: firm + muni. Weights (optional): employment.

Aggregated Firm → Sector Specification

Collapse firm panel to (**sector group** j , **municipality** m , **year** t) cells:

Outcome: share of pre-election base firms with any BNDES

$$H_{jmt}^{\text{pre}} = \frac{1}{N_{jme}^{\text{pre}}} \sum_{f \in \mathcal{F}_{jme}^{\text{pre}}} \mathbf{1}(\text{BNDES}_{fmt} > 0)$$

Instruments: simple average of firm-level FA within each cell

$$\overline{\text{FA}}_{jmt}^{\ell} = \frac{1}{N_{jme}^{\text{pre}}} \sum_{f \in \mathcal{F}_{jme}^{\text{pre}}} \text{FA}_{fmt}^{\ell}$$

Specification:

$$H_{jmt}^{\text{pre}} = \sum_{\ell} \lambda_{\ell} \overline{\text{FA}}_{jmt}^{\ell} + \phi_{jm} + \alpha_{mt} + u_{jmt}$$

FE: muni×sector + muni×year. Clustering: muni + sector. Pre-election base $\mathcal{F}_{jme}^{\text{pre}}$: firms with any non-zero FA instrument. This is the *same* political signal as the firm spec, averaged within cells.

Firm First-Stage: Pooled-Count Exposure (Script 51)

Pending artifact

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Table pending — run script 51 baseline.

- **Unweighted**, extensive-margin, levels specification with firm + muni×year FE
- $\omega_{fp,0}$: share of firm's owners affiliated with party p (pooled count across pre-election period)
- Employment weighted specification: **Mayor** = 0.010* ($F = 3.4$), **Governor** = 0.013*** ($F = 6.9$), **President** = -0.013** ($F = 5.6$)
- Joint M+G+P: $F = 4.8$ on 39.5M observations

Firm First-Stage: Binary Exposure (Script 51)

Pending artifact

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Table pending — run script 51 -exposure=binary.

- $\tilde{\omega}_{fp,0} = \text{mean}(\mathbf{1}(L_{fp} > 0))$: any-party-contact indicator (binary exposure)
- **President** = -0.015^{***} ($F = 7.4$) is the **strongest single-tier instrument**
- Mayor and governor **lose significance** — binary exposure washes out party-share intensity
- Joint M+P: $F = 5.4$; the president signal dominates under binary exposure

Firm First-Stage: Intensive Margin (Script 51)

Exposure	Sample	Mayor	Governor	President	Best F
Pooled-count	BNDES > 0	0.031*	0.029	0.065*	6.0 (P)
Binary	BNDES > 0	0.013	-0.000	0.048*	4.7 (P)

- Intensive-margin sample is much smaller: 472,346 observations, conditional on positive BNDES
- **President dominates** in both exposures; pooled-count is somewhat stronger (0.065*, $F = 6.0$) than binary (0.048*, $F = 4.7$)
- Mayor is marginal only under pooled-count; governor is weak in both cases
- Joint bundles do not improve on the president-only specification: pooled-count M+P $F = 4.1$, binary M+P $F = 2.4$

Script 51, levels specification, coalition alignment, cycle-specific baseline, firm + muni $\times$ year FE, clustered by firm + municipality.

Firm First-Stage: Comparison Across Exposure and Weighting

Exposure	Weighting	Mayor	Governor	President	Best F
Pooled-count	Emp-weighted	0.010*	0.013***	-0.013**	6.9 (G)
Pooled-count	Unweighted	0.002***	0.003***	0.000	38.0 (G)
Binary	Emp-weighted	-	-	-	-
Binary	Unweighted	0.003	0.001	-0.015***	7.4 (P)

- **Pooled-count:** governor dominates; unweighted coefficients are smaller but more precise ($F \gg 10$)
- **Binary:** president dominates; mayor/governor wash out
- **Unweighted:** order-of-magnitude smaller coefficients but much higher F (no employment noise)
- Interaction bundles do not strengthen beyond single-tier baseline ($F = 3.7$)

Key question: Does this firm-level signal survive aggregation to the sector level?

Aggregated First Stage: Pooled-Count Coalition, Unweighted (Script 52)

Pending artifact

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Table pending — run script 52.

- **No coefficient reaches significance** at any conventional level
- President has the largest point estimate (0.010–0.011) with $F = 2.3$ — far from $F = 10$
- Mayor and governor: $F < 1$; joint M+G+P: $F = 1.0$
- Firm-level signal does not survive aggregation

Aggregated First Stage: Binary Coalition, Unweighted

Pending artifact

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```

Table pending — run script 52.

- Binary exposure **recovers significance**: Governor $F = 3.4$, President $F = 5.2$
- Joint M+G+P: $F = 5.0$ — strongest aggregated result across all pooled-count/binary variants
- Binary party alignment even stronger: President $F = 12.9$ (unweighted)

Aggregated First Stage: Pooled-Count Coalition, Emp-Weighted

Pending artifact

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Table pending — run script 52.

- Employment-weighted aggregation further weakens signal: all $F < 1$
- Mayor $F = 1.0$, Governor $F = 0.1$, President $F = 0.3$

Aggregated First Stage: Binary Coalition, Emp-Weighted

Pending artifact

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Table pending — run script 52.

- Binary + emp-weighted: President $F = 8.3$, Governor $F = 3.6$
- Joint M+G+P: $F = 5.6$ — binary exposure partially rescues the signal even under employment weighting

Aggregated First Stage: Full Summary Across Exposure $\times$ Weighting

Exposure	Alignment	Weighting	M F	G F	P F	M+G	M+P	M+G+P
Pooled	Coalition	Unweighted	0.6	0.8	2.3	0.4	1.5	1.0
Pooled	Party	Unweighted	0.0	2.3	0.0	1.2	0.0	0.8
Pooled	Coalition	Emp-wt.	1.0	0.1	0.3	0.7	0.5	0.4
Pooled	Party	Emp-wt.	1.8	0.2	5.7	1.0	4.6	4.1
Binary	Coalition	Unweighted	0.3	3.4	5.2	4.5	4.2	5.0
Binary	Party	Unweighted	0.2	2.8	12.9	1.7	6.7	4.5
Binary	Coalition	Emp-wt.	1.5	3.6	8.3	1.8	6.4	5.6
Binary	Party	Emp-wt.	1.8	2.9	18.8	1.7	9.9	9.5

- **Binary exposure is systematically stronger:** best $F = 18.8$ (binary party, emp-wt., president)
- **President tier dominates** under binary exposure; governor under pooled-count
- Pooled-count instruments are uniformly weak at the aggregated level ($F < 5.7$)
- Bold values exceed Stock-Yogo $F = 5$ threshold; binary party emp-weighted exceeds $F = 10$

Aggregated First Stage: Party Alignment (Unweighted)

Pending artifact

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Table pending — run script 52.

- Governor: largest point estimate (0.005–0.006, $F = 2.3$)
- Mayor: essentially zero (0.001, $F = 0.0$); President: negative but tiny (-0.002 , $F = 0.0$)
- Party alignment performs similarly poorly to coalition at the aggregated level (pooled-count)

Comparison: Firm vs. Aggregated First Stage

Tier	Pooled-count exposure				Binary exposure			
	Firm		Agg.		Firm		Agg.	
	Coef	F	Coef	F	Coef	F	Coef	F
Mayor	0.010*	3.4	0.003	0.6	0.003	0.8	—	0.3
Governor	0.013***	6.9	0.004	0.8	0.001	0.1	—	3.4
President	-0.013**	5.6	0.010	2.3	-0.015***	7.4	—	5.2
M+G+P	—	4.8	—	1.0	—	4.1	—	5.0
N	39.5M		293K		39.5M		293K	

- **Pooled-count:** firm signal attenuates 70–90% at aggregation; president sign flips
- **Binary:** president signal **survives** aggregation ($F = 5.2$); joint M+G+P even improves ($F = 5.0$)
- Binary exposure is less noisy in thin cells because it does not depend on party-share intensity

Why does pooled-count signal disappear? → systematic diagnostics follow

Diagnostic #1: Cell Sparsity (Dominant Driver)

Most (sector, muni, year) cells have very few pre-election firms:

Firms/cell	Cells	Share
1	111,385	36.7%
2	49,584	16.3%
3–5	62,502	20.6%
6–10	34,493	11.4%
11–50	37,826	12.5%
50+	7,644	2.5%

Restricting to dense cells recovers the signal:

Min N	Cells	Coal. F	Party F
All	293K	1.02	0.84
≥ 5	74K	0.97	1.05
≥ 10	35K	1.38	0.89
≥ 20	15.5K	2.96	1.35
≥ 50	4.6K	6.89	0.77

- 53% of cells have ≤ 2 pre-election firms — H_{jmt}^{pre} takes values $\{0, 0.5, 1\}$ for half the sample
- Coalition F monotonically rises to **6.89** at $N \geq 50$ — but only 2.5% of cells survive
- The firm signal *is there* in dense cells; thin cells destroy it

Diagnostic #2: Outcome Homogeneity Within Muni-Years

Even if instruments varied across sectors, the **outcome itself does not**:

Statistic	Value
Median within-muni-year SD of H_{jmt}^{pre} across sectors	0.000
Median within-muni-year range of H_{jmt}^{pre}	0.000
75th percentile within-muni-year SD	0.061
Muni-years with zero positive sector-group cells	42.3%
Muni-years with exactly one positive sector-group cell	20.4%
Muni-years with ≤ 1 positive sector	62.7%
Mean share of active groups with positive BNDES	17.4%
Median HHI of grouped BNDES shares (positive m.y.)	0.670

- In most muni-years, **all sectors have the same BNDES outcome** (all zero or all positive)
- 62.7% of muni-years have ≤ 1 sector with positive BNDES — **no cross-sector variation to explain**
- Where BNDES is positive, it is highly concentrated: $\text{HHI} = 0.67$, top sector gets 79.9%

Diagnostic #3: Limited Instrument Variation Across Sectors

Within-muni-year instrument variation

Instrument	Med. SD	FE R^2
$\overline{FA}_{coal.}^M$	0.224	0.718
$\overline{FA}_{coal.}^G$	0.205	0.760
$\overline{FA}_{coal.}^P$	0.142	0.737
$\overline{FA}_{party}^M$	0.064	0.708
$\overline{FA}_{party}^G$	0.057	0.709
$\overline{FA}_{party}^P$	0.059	0.731

Alternative FE specifications

FE structure	Coal. F	Party F
$m \times j + m \times t$	1.02	0.84
$m \times j + j \times t$	0.51	0.56
$j + t$ (minimal)	1.56	1.58
$m \times j$ only	0.98	3.79

- Party instruments have very little cross-sector spread ($SD \approx 0.06$)
- FE absorb 71–76% of instrument variation — but this is **not the binding constraint**
- No FE combination recovers a strong first stage — the problem is **structural**

Diagnostic #4: Sector Heterogeneity — Two Sectors Carry All Signal

Sector	Own F	Leave-one-out F	Interpretation
Tp (Transport)	20.4	0.37	Dropping it kills pooled signal
XX (Residual)	47.6	0.97	Dropping it kills pooled signal
CA (Adv. Manuf.)	3.2	5.2	Dropping it <i>raises</i> pooled F
UCo (Utilities)	2.7	1.2	
Tr (Trade)	2.2	1.7	
MS (Market Svcs.)	1.4	1.2	
PSO (Public/Social)	0.4	1.0	
Ag (Agriculture)	0.7	1.0	
Mi (Mining)	0.1	1.0	

- Only **Transport** ($F = 20.4$) and **Residual** ($F = 47.6$) show strong instruments
- CA (Advanced Manufacturing) **actively dilutes** the pooled signal: its inclusion drops F from 5.2 to 1.02
- The remaining 6 sectors contribute pure noise ($F < 2.2$)

Diagnostic Summary: Why Aggregation Fails

Rank	Hypothesis	Key evidence	Score
1	Cell sparsity	53% of cells have ≤ 2 firms; F rises 1.0 $\rightarrow$ 6.9 at $N \geq 50$	5.87
2	FE absorption	FE absorb 73% of instrument var.; even minimal FE give $F < 1.6$	0.74
3	Diffuse exposure	Median party-sector HHI = 0.59; within-m.y. outcome SD = 0	0.41
4	Aggregation form	Owner-weighted $F = 0.83$ (worse); raw vs. weighted $\rho = 0.94$	-0.18

Bottom line — the collapse is driven by three structural features:

- ① **Thin cells:** most cells have 1–2 firms $\Rightarrow$ cell averages are extremely noisy
- ② **Homogeneous outcomes:** 62.7% of muni-years have ≤ 1 sector with positive BNDES $\Rightarrow$ no cross-sector variation to explain
- ③ **Concentrated signal:** only Transport and Residual show instrument relevance; the other 7 sectors contribute noise

Sector First Stage: Baseline (Script 53)

Pending artifact

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Table pending — run script 53 with `-specs=baseline`.

- Sector-level shift-share instruments Z_{jmt}^{ℓ} (alignment $\times$ baseline sector-party exposure weights)
- FE: `muni` $\times$ `sector` + `sector` $\times$ `year`; clustering: `muni` + `sector`
- Owner-count, coalition, cycle-specific baseline — all $F < 6$

Sector First Stage: Full Specification Battery (Script 53)

Instrument weight	Alignment	FE	Controls	F range	Best F
<i>m×j + j×t FE (primary)</i>					
Owner-count	Coalition	j×t	No	0.0–5.9	5.9
Owner-count	Party	j×t	No	0.0–0.1	0.1
Employment	Coalition	j×t	No	2.8–8.3	8.3
Employment	Party	j×t	No	1.0–1.7	1.7
Equal-firm	Coalition	j×t	No	3.3–10.7	10.7
Equal-firm	Party	j×t	No	0.0–2.1	2.1
Binary	Coalition	j×t	No	1.0–3.4	3.4
Binary	Party	j×t	No	0.1–0.9	0.9
<i>m×j + m×t FE (robustness)</i>					
Owner-count	Coalition	m×t	No	0.0–2.3	2.3
Owner-count	Party	m×t	No	0.0–1.4	1.4
Employment	Coalition	m×t	No	0.8–2.3	2.3
Employment	Party	m×t	No	0.5–1.5	1.5
Equal-firm	Coalition	m×t	No	1.2–2.8	2.8
Equal-firm	Party	m×t	No	0.1–1.5	1.5
Binary	Coalition	m×t	No	0.0–2.0	2.0
Binary	Party	m×t	No	0.5–2.9	2.9

$N = 795,456$. Levels specification (s_{mjt}), cycle-specific baseline. F : Wald test on all Z instruments jointly. Best F = max across 6 tier combos (M, G, P, M+G, M+P, All). Bold $\geq$ Stock-Yogo 5% critical value.

Sector First Stage: Key Findings Across Specifications

- **Equal-firm, coalition, $j \times t$ FE**: best single-combo $F = 10.7$ — the only specification exceeding Stock-Yogo 10
- **Employment-weighted**: second strongest ($F_{\max} = 8.3$); approaching relevance
- **Owner-count**: weak under coalition ($F \leq 5.9$), near-zero under party alignment
- **Binary**: weakest weight variant at sector level ($F \leq 3.4$) — opposite of firm-level pattern
- **Party alignment uniformly weak**: $F < 3$ across all weight variants and FE choices
- **$m \times t$ FE**: reduces all F relative to $j \times t$ FE (absorbs muni-year variation)

Interpretation: Equal-firm and employment weights concentrate exposure on municipalities where BNDES lending actually varies across sectors. Coalition alignment is the operative margin at the sector level. The sector pipeline shows **some promise under specific weight/FE combinations**, unlike the uniformly weak aggregated firm→sector route.

Why the Current Sector Grouping Fails

The current 10 sector groups (from 21 CNAE sections) were designed for broad economic categories, **not** for BNDES lending patterns:

Issue	Value	Implication
Median firms per cell (overall)	6	Thin cells
Median pre-election firms per cell	2	Very thin instrument base
Cells with 0 affiliated firms	32.1%	No instrument variation
Muni-years with ≤ 1 pos. sector	62.7%	No cross-sector outcome var.
HHI of BNDES shares (pos. m.y.)	0.670	Lending highly concentrated
Top-group share (pos. m.y.)	79.9%	Nearly all BNDES in one sector
Mean affiliation: Tp (Transport)	38.4%	Instrument support concentrated
Mean affiliation: Ag (Agriculture)	4.8%	Near-zero instrument support

The sector grouping creates cells that are simultaneously too sparse for reliable aggregation and too fine-grained relative to the coarseness of BNDES lending patterns.

Alternative Paths Forward

Option A: Coarser sector groupings aligned with BNDES lending

- Define 3–5 “BNDES-relevant” sectors based on where lending occurs and affiliation support exists
- Evidence: Transport and Residual carry the signal — perhaps a Manufacturing / Services / Transport+Other trichotomy
- **Pro**: denser cells, more outcome variation, preserves sector story
- **Con**: fewer sectors $\Rightarrow$ less heterogeneity for second stage

Option B: Firm-level reallocation summary variable

- Construct a municipality-level measure of *firm-level* BNDES reallocation (e.g., turnover in firm access between $t-1$ and t , or firm-level HHI within municipality)
- Instrument with municipality-level alignment — **bypass** the sector dimension
- **Pro**: uses the strong firm-level signal directly; no sector-aggregation noise
- **Con**: new theoretical motivation needed; less interpretable for sector misallocation

Option C: Return to ungrouped sectors (21 CNAE sections)

- Ungrouped with muni $\times$ year FE achieves $F = 12.4$ for mayor (above Stock-Yogo 10)
- More sections $\rightarrow$ more variation in shares and instruments

Conclusion & Next Steps

Firm-level validation is solid: political alignment $\rightarrow$ BNDES access is a real channel.

Binary exposure is the key finding:

- **Firm level:** binary president $F = 7.4$ vs. pooled-count governor $F = 6.9$
- **Aggregated:** binary survives (F up to 18.8) while pooled-count collapses ($F < 2.3$)
- **Sector shift-share:** equal-firm weights reach $F = 10.7$; employment $F = 8.3$

Remaining challenges: 53% of cells have ≤ 2 firms; 63% of muni-years have ≤ 1 positive sector; party alignment uniformly weak at sector level.

Next steps:

- ➊ Pursue **equal-firm / employment sector instruments** with $j \times t$ FE as the primary sector specification
- ➋ Investigate the **binary aggregated route** (binary party, emp-weighted, $F = 18.8$) as an alternative pathway
- ➌ Run ungrouped sector first stage (21 CNAE sections) — earlier result suggested $F = 12.4$ for mayor
- ➍ Evaluate evidence and choose the path to the second stage (CDB)

Appendix A: Aggregation Mechanics

Pre-election firm base $\mathcal{F}_{jme}^{\text{pre}}$:

- All firm-muni-year rows where at least one $\text{FA}_{fmt}^{\ell} \neq 0$
- 2.47M rows out of 44.2M (5.6% of full firm panel)
- Collapsed to 303K (sector $\times$ muni $\times$ year) cells

Aggregation: simple (unweighted) average within each cell

- $H_{jmt}^{\text{pre}} = \text{mean}(\mathbf{1}(\text{BNDES}_{fmt} > 0))$ over pre-election firms
- $\overline{\text{FA}}_{jmt}^{\ell} = \text{mean}(\text{FA}_{fmt}^{\ell})$ over pre-election firms
- No employment weighting at the aggregation stage

FE and clustering:

- FE: muni $\times$ sector group + muni $\times$ year (8,606 FE groups; 2,049 singletons removed)
- Two-way clustering: municipality + sector group
- Effective sample: 292,926 observations after singleton removal

Appendix B: Sector Group Definitions

Group	Label	CNAE Sections
Ag	Agriculture	A
Mi	Mining	B
CL	Light Manufacturing	C (div. 10–18: food, textiles, wood, paper, printing)
CH	Heavy Manufacturing	C (div. 19–25: petroleum, chemicals, metals, rubber)
CA	Advanced Manufacturing	C (div. 26–33: electronics, machinery, vehicles)
UCo	Utilities & Construction	D (electricity/gas), E (water/waste), F (construction)
Tr	Trade	G (wholesale and retail)
Tp	Transport	H (transport and storage)
MS	Market Services	I, J, L, M–N (accommodation, info, real estate, prof.)
PSO	Public, Social & Other	P (education), Q (health), R (arts), S (other services)

Appendix C: Aggregated Panel Descriptive Statistics

	Firm panel	Aggregated panel
Observations	44,181,405	303,434
Pre-election base	2,467,350	303,434
Unique firms	455,356	—
Unique munis	5,557	5,557
Sector groups	9	9
H_{jmt}^{pre} : mean	—	0.032
H_{jmt}^{pre} : sd	—	0.128
Median N^{pre} per cell	—	2
Max N^{pre} per cell	—	2,891

- **Sparse:** mean $H^{\text{pre}} = 3.2\%$ — only 1 in 30 pre-election firms receive BNDES
- **Thin cells:** median 2 firms per (sector, muni, year) cell
- 2,049 singleton FE removed (10,508 observations dropped)

Appendix D: Aggregated First-Stage Summary Table

Tier	Coalition alignment					
	M	G	P	M+G	M+P	M+G+P
Coef	0.003	0.004	0.010	—	—	—
SE	(0.004)	(0.004)	(0.007)	—	—	—
<i>F</i>	0.6	0.8	2.3	0.4	1.5	1.0
	Party alignment					
	M	G	P	M+G	M+P	M+G+P
Coef	0.001	0.005	-0.002	—	—	—
SE	(0.004)	(0.003)	(0.014)	—	—	—
<i>F</i>	0.0	2.3	0.0	1.2	—	0.8

- **All** $F < 2.3$ — no instrument combination approaches relevance
- 293K observations (after singleton removal) with $\text{muni} \times \text{sector} + \text{muni} \times \text{year}$ FE

Appendix E: Cell Size Distribution by Sector Group



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Cell size distributions vary substantially across sectors. Transport (Tp) and Trade (Tr) have relatively denser cells; Agriculture (Ag) and Mining (Mi) are dominated by singletons.

Appendix F: BNDES Positive Share by Sector × Year



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Share of cells with positive BNDES credit by sector group and year. Trade (Tr) consistently has the highest positive-cell rate.

Appendix G: Political Affiliation Support by Sector × Year



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Mean affiliation share (fraction of firms with any politically affiliated owner) by sector group and year. Transport (Tp) has the highest support; Agriculture (Ag) the lowest.